UTKARSH SINGH

B.Tech CSE & AI
Rishihood University

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•SUMMARY

CS & AI undergraduate, *cybersecurity by choice*. Explores **web and API vulnerabilities**, sharpens security instincts through **CTF competitions**, and documents reproducible findings for responsible verification.

•EDUCATION

• Rishihood University **SGPA: 8.09**
B.Tech in Computer Science and Artificial Intelligence
2025 – 2029

•EXPERIENCE

• Malaviya National Institute of Technology Jaipur **Jul 2026 – Present**
Application Security Intern
- Working on **Android security testing**, analyzing Android application components and platform-level components for potential security issues.
- Performing **kernel fuzzing** and security testing of native and userspace libraries to identify crashes, vulnerabilities, and unexpected behavior.
- Working with the **Android HAL** and other Android platform components, conducting security-focused testing across different layers of the Android stack.

• Cyberops Infosec LLP **Jun 2026 – Aug 2026**
Web Security Intern
- Worked on **web application security testing**, security audits, and black-box audits.
- Performed security assessments based on the **OWASP Top 10** and SANS Top 25, focusing on identifying and validating common web vulnerabilities.
- Worked with **honeypot technologies** and explored offensive security techniques as part of practical security testing.

• Amdox Technologies, Anekal, Karnataka **Nov 2025 – Jan 2026**
Web Developer
- Developed **Rust-based web applications** and backend services using Actix Web, focusing on modular and maintainable server-side architecture.
- Built and integrated **REST APIs**, handling request validation, routing, data processing, and structured error handling.
- Worked on application reliability through **defensive programming**, reusable components, and robust API handling.

•PROJECTS

• Whispr — Client-Side Encrypted Messaging Platform [GitHub]
Next.js, Web Crypto API, Supabase Auth, PostgreSQL, Realtime
- Implemented browser-side **ECDH P-256** key agreement and **AES-GCM** authenticated encryption with ciphertext-only message persistence.
- Designed **local keyring management**, encrypted private-key backups, key-version tracking, and participant-scoped database access.

• SecuScan — Local-First Security Scanning Workspace [GitHub]
FastAPI, React, TypeScript, Docker, PostgreSQL/SQLite, Redis
- Designed a **metadata-driven plugin system** with safety classification, consent-aware tasks, integrity checks, audit events, and normalized findings.
- Unified capability, network, rate, timeout, and concurrency controls across CLI, Python, and Docker-oriented scan execution.
- Structured scan output for **severity-based review**, evidence collection, report generation, and manual validation of automated findings.

• Open World Game Engine [GitHub]
C++17, SDL2, Metal 3.0, CMake
- Built a modular open-world game engine with a **Metal-based 3D renderer** featuring PBR lighting, skybox rendering, and frustum culling.
- Implemented **custom physics**, FPS/TPS character controllers, vehicle physics, and gameplay systems for movement and interaction.
- Added **chunk-based world streaming** with distance-based LOD, A* pathfinding, behavior-driven NPCs, dynamic traffic, missions, inventory, weapons, and binary save/load systems.

•SKILLS

•Security
Android Security · Android Application Components · Android HAL · Kernel Fuzzing · Native & Userspace Fuzzing · Web Security · OWASP Top 10 · SANS Top 25 · VAPT · Offensive Security · Vulnerability Assessment

•Languages
C · Rust · Python · JavaScript · TypeScript · Bash/Shell

•Frameworks
Actix Web · Axum · FastAPI · Node.js · Express · React · Next.js

•Databases
PostgreSQL · MariaDB · SQLite · Redis · Neo4j · Qdrant

•Tools
Syzkaller · ADB · GDB · LLDB · Frida · Ghidra · IDA Pro · Burp Suite · Metasploit · Nmap · Wireshark · Nuclei · ffuf · Gobuster · OWASP ZAP · T-Pot · Cowrie · Dionaea · OpenCanary · Docker · Git · Linux

•Concepts
Computer Networking · REST APIs · API Security · Application Security · Fuzzing · Authentication · Authorization · Session Security · Business Logic Testing · Security Reporting

•CERTIFICATES

- OWASP Top 10 — **Packt** (Jul 2026)
- Advanced Cybersecurity Techniques — **Johns Hopkins** (Jun 2026)
- Rust Programming and DevOps — **Duke University** (Jun 2026)
- Certified in Cybersecurity (CC) — **ISC2** (May 2026)
- Network & Network Security — **Google** (Apr 2026)
- Web Application Security — **Microsoft** (Apr 2026)
- AWS Cloud Security Foundations — **AWS** (Mar 2026)

•ACHIEVEMENTS

- Project Admin & Open-Source Contributor, GirlScript Summer of Code (GSSoC) 2025 – 2026
- Top 10 in Various National and International Cybersecurity Competitions